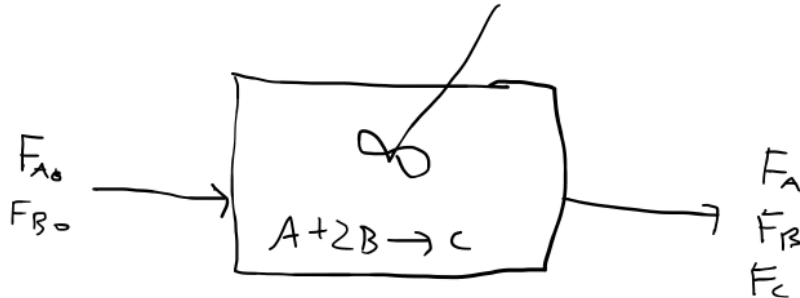


EOX Practicing with Stoichiometric Tables and Changes in Volumetric Flowrate

The gas phase reaction $A + 2B \rightarrow C$ is carried out isothermally and isobarically in a fluidized CSTR with no pressure drop. The feed is stoichiometric in A and B and no C is fed. The pressure, temperature, and volumetric flowrate of the feed gas are 6 atm, 170°C, and 50 L/min respectively. The rate constant $k_A = 53 \text{ mol/kg catalyst} \cdot \text{min} \cdot \text{atm}^3$ at 300 K, and the activation energy is 80 kJ/mol. Complete the stoichiometric table and write the rate as a function of conversion.



Species	initial	change	exiting
A	$F_{A0} = \frac{Y_{A0} P_0 v_0}{R T_0}$	$-F_{A0} X_A$	$F_A = F_{A0} (1 - X_A)$
B	$F_{B0} = 2 F_{A0}$	$-F_{B0} X_B = -2 F_{A0} X_A$	$F_B = 2 F_A = 2 F_{A0} (1 - X_A)$
C	0	$F_{A0} X_A$	$F_C = F_{A0} X_A$

$$F_{A0} = \frac{\left(\frac{1}{3}\right)(6 \text{ atm})\left(50 \frac{\text{L}}{\text{min}}\right)}{\left(0.0821 \frac{\text{Latm}}{\text{molK}}\right)(170^\circ\text{C} + 273.15\text{K})}$$

$$Y_{A0} = \frac{1 \text{ mol A}}{1 \text{ mol A} + 2 \text{ mol B}} = \frac{1}{3}$$

- assume elementary kinetics

- $-r_A = k_A C_A C_B^2$. for fluidized CSTR, the units on the rate should be equal to $\frac{\text{mol}}{\text{time} \cdot \text{mass catalyst}}$. This is because

the mole balance for a fluidized CSTR is $W = \frac{F_{A0} X_A}{-r_A}$ instead of $V = \frac{F_{A0} X_A}{-r_A}$

for a "normal" CSTR.

- so rate is: $-r_A = k_A P_A P_B^2$. Recalling that

$$C_i = \frac{P_i}{RT} = \frac{y_i P_{\text{tot}}}{RT}$$

$$-r_A = k_A P_A P_B^2$$

Eq 4-25 or Eq 4-27

$$P_A(X_A) = RT C_A(X_A)$$

$$P_i = \frac{P_{A0} (\Theta_i + v_i X_A)}{1 + \epsilon X_A} \quad p \text{ here} = 1 \text{ since the reactor is isobaric}$$

$$\epsilon = Y_{A0} S, \quad S = \left(\frac{d}{a} + \frac{c}{a} - \frac{b}{a} - 1 \right) = -2$$

$$X \quad \frac{1}{1} - \frac{2}{1} - 1$$

$$\epsilon = -\frac{2}{3}$$

$$P_A = \frac{P_{A0} (1 - X_A)}{1 - \frac{2X_A}{3}}, \quad P_B = 2P_A$$

$$P_{A0} = Y_{A0} P_0$$